AYUSH SINGH

SUMMARY

Software Engineer III (L4) at Google (Munich) with over **4 years** of experience architecting planet-scale distributed systems, low-latency **C++, Java, and Go** microservices, fault-tolerant storage systems, and real-time streaming pipelines. Standardized production architectures across **5,200+ server nodes**, scaled internal storage backing **100,000+ services**, and delivered **$2.5M+ in quarterly savings**. Earned a fast-track promotion to L4 in 15 months, a **VP of Engineering Spot Bonus**, and **3 Google Readability certifications** across **1,040+ code reviews**.

SKILLS

Languages

C++ · Java · Go · TypeScript · Python · Scala · Kotlin · SQL

Distributed Systems & RPC

gRPC · Protobuf · HDFS / FUSE · Paxos / Raft · Apache Spark · Kafka · ActiveMQ

Storage, Cloud & Observability

Spanner · Snowflake · Redis · PostgreSQL · Kubernetes · Docker · OpenTelemetry · GCP / AWS · Terraform

AWARDS

- **VP of Engineering Spot Bonus** & **7 Peer Bonuses** at Google for global launches and storage reliability.
- **3 Google Readability Certifications** in C++, TypeScript, and API Design (**1,040+ code reviews**).
- **Rank #1 in Institute & #72 Worldwide** on GeeksforGeeks (top **0.07%** of 100,000+ developers).
- **Indian Academy of Sciences Fellow** (**1 of 11** nationwide) & **Google ML Facilitator** (**1 of 50**).

FIND ME ONLINE

GitHub
https://github.com/iu-sh

Portfolio & Interactive Resume
https://ayushsingh.is-a.dev

LinkedIn
https://linkedin.com/in/iu-sh

Software Engineer III (L4) · Distributed Systems & Core Infrastructure

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EXPERIENCE

Software Engineer III (L4) — Core Infrastructure & Distributed Systems
11/2024 - Present

Google — Munich, Germany

- Architected high-throughput, low-latency request processors in Google's core **Java backend framework** and **C++ Search engine**, sustaining **7 billion+ daily requests** across Search, YouTube, and Maps.
- Standardized the Google wide SignedOut request-processing architecture defaulted across **5,200+ server nodes** and **210+ production modules**, cutting signal-processing complexity by **8%** and retiring **1,300+ lines of code**.
- Implemented a parallel stream shadow-diffing and replay validation pipeline across **YouTube, Play, Search, and Payments**, verifying **100% live migration parity** across **10 billion+ daily RPCs** with zero regressions.
- Migrated **7 critical identity and policy evaluation pipelines** from synchronous request paths to distributed batch compute, reducing p99 serving tail latency by **12%** and reclaiming GCUs.
- Spearheaded distributed backend launches for **European Online Safety** and **US App Store regulatory compliance** across Web, Android, and iOS in **4 weeks** across **5 engineering teams** (**VP of Engineering Spot Bonus**).
- Built an autonomous **multi-agent Python diagnostic system** fanning out parallel queries across rollout logs, code changes, and metrics to email root-cause reports before on-call triage; **3x Google Readability** (**1,040+ code reviews**).

Software Engineer II → III (L4) — Distributed Storage & Observability
07/2022 - 11/2024

Google — Hyderabad, India

- Engineered fault-tolerant storage and replication components for Google's internal **HDFS-like distributed file system**, accelerating binary delivery across **100,000+ production services**.
- Deployed server-side admission control, bounded semaphore RPC concurrency limits in the **FUSE client daemon** to prevent cascading OOMs, and decentralized **Paxos snapshot locks**, cutting lock contention by **65%**.
- Designed an automated **SLA/SLO compliance and observability platform** correlating distributed traces and metrics across **40+ product teams**, eliminating manual audits and generating **$2.5M+ in quarterly savings**.
- Co-developed a real-time **OpenTelemetry** trace collector over gRPC and an encrypted **Go asset gateway** sustaining **50,000+ requests per second** with zero-downtime key rotation.
- Led technical readiness and dependency analysis for Chrome third-party cookie deprecation across **120+ enterprise services** (**approved by Google CEO**) and earned a **fast-track promotion to L4 in 15 months**.

INTERNSHIPS

- **Komprise** *(Software Engineer Intern · 01/2022 - 06/2022 · Bengaluru):* Automated parallel multi-threaded cluster verification across storage nodes, cutting build verification runtime by **87.5%** and increasing bug detection coverage by **60%**.
- **Accenture** *(Data Engineer Intern · 05/2021 - 07/2021 · Bengaluru):* Constructed distributed cloud ETL pipelines transferring **10TB+ datasets** from **AWS S3 to Snowflake**, reducing manual analytics reporting time by **70%**.
- **Tookitaki** *(Software Engineer Intern · 10/2020 - 01/2021 · Singapore):* Optimized a real-time **Apache Spark Structured Streaming and ActiveMQ** pipeline with connection pooling, reducing end-to-end latency by **94%** (**850ms to 45ms**).

PROJECTS

YTM AI Host (Real-Time AI Radio Jockey)
github.com/iu-sh/ytm_ai_host

TypeScript, Python, Gemini API, WebGPU

- Developed a local audio streaming server and browser extension for YouTube Music and Spotify combining hardware-accelerated GPU inference and neural TTS for **sub-100ms audio transitions** across **1,000+ tracks**.

LinkinJMS (Apache Spark ActiveMQ Connector)
github.com/iu-sh/linkinJMS

Scala, Apache Spark, ActiveMQ, AMQP/TLS

- Created an open-source bidirectional ActiveMQ streaming connector for Apache Spark with exactly-once delivery and encrypted transport, sustaining **sub-50ms ingestion latency** across **100,000+ messages/sec**.

EDUCATION

Bachelor of Technology in Electrical Engineering
08/2018 - 05/2022

National Institute of Technology, Agartala — CGPA: 9.32 / 10.0